(Humidification)

Pre-Lab Plan
Unit Operations Lab #4
Date: 10/15/2025

Group 2, Tuesday, 1 PM section
Team Members:
Cesearo Dilosa
Javier Silva
Ameen Abuzir
Joshua Laforest

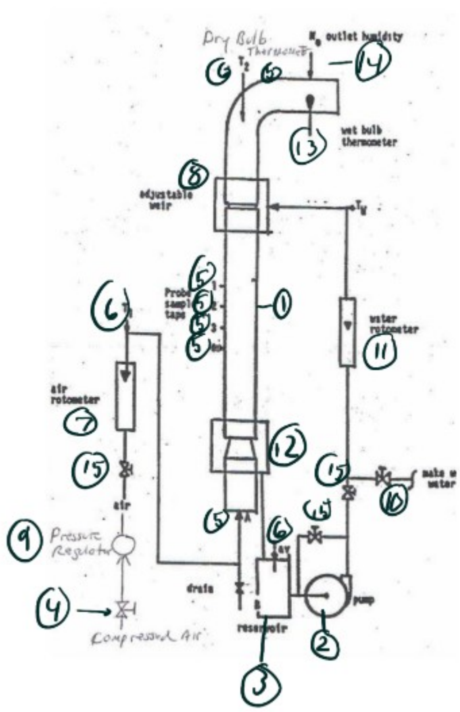
1. Experimental Objective (5 pts)

The objective of the experiment is to determine the heat and mass transfer coefficients for the humidification of air through a wetted-wall column. Flowrate and air temperature are varied to analyze the different operating conditions that affect the transfer of mass and heat between deionized water and air. Humidity and temperature data are then collected to calculate the transport coefficients and identify any observable trends.

2. Experimental

2.1 Apparatus (5pts)

Apparatus Figure:



~Figure 1: Apparatus with components listed numerically.

Apparatus Components

COMPONENT NO.	EQUIPMENT	DESCRIPTION
1	Humidification Column	Used to pass water and air to humidify the air in the system.
2	Centrifugal Pump	Pump water from the reservoir tank down the sides of the column. Contains a pump switch for operation.
3	Water Tank	Holds water.
4	Compressed Air	Air that will flow throughout the system.
5	Thermocouples	Used to measure the temperature at different points of the apparatus.
6	Thermometer	Used to measure the temperature in the inlet.
7	Air Flow Meter	Used to measure air flow in the system.
8	Weir	At the top of the column; ensures water is directed to the sides of the column.
9	Pressure Regulator	Controls outlet pressure of compressed air.
10	Pump Release Valve	Used to empty the water out of the system.
11	Water Flow Meter	Measures water going in the system.
12	Heat Exchanger	Heat incoming air.
13	Wet Bulb Thermometer	Measure the temperature of the moisturized air.
14	Hygrometer	Determine the humidity of air leaving the system.
15	Valves	Manipulation of the various valves in the system directs the flow of water, steam, and air.

2.2 Materials and Supplies (5 pts)

MATERIALS & SUPPLIES	DESCRIPTION
Anemometer	Device used to record the humidity and temperature
5L plastic bucket	Used to house the deionized water
Measuring tape	Used to measure the length of the wetted wall section
Heat Gloves	Used to safely manipulate the steam valve for the heat exchanger
Timer	Used to track the time when the anemometer reaches steady state which typically is within 20 minutes

2.3 Experimental Plan (10 pts)

PROCEDURE:

1. Begin by flowing water at a sufficient rate through the tube to completely wet the inside walls of the cylinder. Do this without air.
2. Next, establish an airflow and let the system reach steady state.
3. Once steady state is achieved, record water and air temperatures at various locations. Record humidity levels as well.
4. Record data at the same three air flowrates as last week, but this time use steam to vary the temperature of the air.
5. Complete three sets of measurements at three different air temperatures.
6. Analyze and interpret data.

Project Safety Evaluation (10 pts)

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: Wires and electrical equipment are present
Potential mechanical hazards?	NO	If yes, identify conditions:
Potential pressure hazards?	Yes	If yes, identify conditions: Pressure build up with the air flow and steam device.
Potential temperature hazards?	Yes	If yes, identify conditions: Reboiler is present for hot steam. Pipes, tanks
Hazardous/toxic chemicals involved?	NO	If yes, hazards are involved:
Gloves required?	Yes	If yes, specify type: Thermal resistant gloves.
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: CAS #7732-18-2 Steam.
Process Parameters	*Max* *Expected*	*List location and possible hazards and precautions*
Temperature (ºC)	100 C	At the reboiler and pipes holding in steam and connecting reboiler to heat exchanger.
Pressure (psia)	~40 psia	System pressure above this value isn't needed.

Safety Controls	*yes/no*	*If yes, Provide description*
Pressure Regulator	Yes	Controls the amount of air flow in the system.
Thermal gloves	Yes	To work by controlling the amount of steam flowing and measuring hot air leaving.
Goggles	Yes	Pressure and hot steam are present when conducting the lab.
Steam trap	Yes	Prevents the steam from leaking.
Pump Procedure	Yes	A procedure to prevent the pump from being damaged. Turn on the pump for 3 seconds, shut off the pump for 10 seconds, and turn back on the pump.

2.4 References:

1. *"Humidification Lab Manual." ChE 382: Unit Operations Laboratory*, Rev. Spring 2018v1, Department of Chemical Engineering, [University of Illinois at Chicago]. PDF.

Safety Statement

I have read relevant background material for the Unit Operations Laboratory entitled: “Humidification” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance with standards and acceptable safety practices and will conduct all my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Team Member Printed Signature: Cesearo Dilosa   Date: 15 October 2025

Team Member Printed Signature: Joshua Laforest   Date: 15 October 2025

Team Member Printed Signature: Javier Silva   Date: 15 October 2025

Team Member Printed Signature: Ameen Abuzir   Date: 15 October 2025